

Reference	Fill Material Location	Material Description	AASHTO Designation	Compaction Requirements
④	Final backfill / material to grade	For backfill, the design engineer will specify suitable native or engineered materials to create a stable subgrade as outlined in the project plans. Recommended material should be at minimum 12" of clean, well-graded soil.	Not applicable (N/A)	Subgrade preparation will follow strict specifications on the site plans, which may include additional requirements for paved installations as determined by the site engineer. The degree of soil compaction should correspond with existing soil and water conditions and future external load.
③	Initial backfill / cover over system	Granular well-graded soil/aggregate mixtures, < 35% fines, most pavement subbase materials can be used in this layer.	AASHTO M145 A-1, A-2-4, A-3 AASHTO M43 3, 357, 4, 467, 5, 56, 57, 6, 67, 68, 7, 78, 8, 9, 10	Compaction begins after 12" of material placement over the system, with subsequent lifts no greater than 12" for well-graded materials of 6" for processed aggregates, all compacted to a minimum of 95% standard Proctor density (SP)
②	Embedment / perimeter material	Clean, crushed angular stone; Native soil or sand compacted per application requirements.	AASHTO M43 3, 357, 4, 467, 5, 56, 57	Bedding must be uniform, level and free of debris, at a minimum depth of 4" and compacted to a minimum density of at least 95% standard Proctor density (SP), or as required by engineer.
①	Foundation / bedding material			

Select backfill

Finished Surface

Inspection / Access Port

④

Geotextile wrapped around tank and backfill (based on application)

③

Excavation wall (Sloped or vertical)

②

①

Undisturbed Sub-base*

Inspection / Access Port

Select backfill

Finished Surface

Geotextile wrapped around tank (based on application)

④

③

②

Excavation wall (Sloped or vertical)

①

Undisturbed Sub-base*

*Sub-base must provide a minimum bearing capacity of 2,000 lbs/ft² prior to installing foundation materials and/or AquaCell system

Notes:

A. Reference **Installation Min. Loading & Depth Requirements (AQ-100-02)** for load, compaction, and min/max depth requirements.

B. Reference **Geosynthetics Detail (AQ-100-03)** for minimum fabric material weights and application details.

C. Reference **AquaCell Installation Guide** for further information and instruction on installation of backfill material.

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Title: AquaCell Acceptable Backfill Materials Table

Date: 10/28/2025 Scale: NTS Drawing: AQ-100-01 Rev: 4

Drawn: HJD Approved: JS Sheet: 1 of 1

Load Rating ^A	Min. / Max. Top Cover (SC) ^B	Min. / Max. Top Cover (EX) ^C	Soil Compaction (SP) Minimum ^D	Compacted Backfill Lifts Min. / Max. ^E
Light Duty < 2,500 lbs	12 inches / 14.4 feet	12 inches / 26.2 feet	90% SP (Standard Proctor)	6 in. /12 in.
H-10 Traffic Loading	12 inches / 14.4 feet	12 inches / 26.2 feet	95% SP (Standard Proctor)	6 in. /12 in.
HS-20 Traffic Loading	18 inches ¹ / 14.4 feet	16 inches ² / 26.2 feet	95% SP (Standard Proctor)	6 in. /12 in.
HS-25 Traffic Loading	30 inches / 14.1 feet	22 inches / 26.1 feet	98% SP (Standard Proctor)	6 in. /12 in.

Inspection/Access Port

Select Compacted Backfill ^G

HS-20/25 Finished Surface

Compacted Backfill ^G

Excavation Wall (Sloped or vertical)

Undisturbed Sub-base*

Concrete collar for HS-20/25 loads (required & specified by E.O.R)

Geotextile (based on application)

⑩12"/18"/30"

⑩12"/16"/22"

AQUACELL SC (STANDARD)

4" Foundation Min.

Inspection/Access Port

Select Compacted Backfill ^G

Non-Traffic Surface

Compacted Backfill ^G

Excavation wall (Sloped or vertical)

Undisturbed Sub-base*

Geotextile (based on application)

⑩12" (NTS)

⑩12"

AQUACELL SC (STANDARD)

4" Foundation Min.

*Sub-base must provide a minimum bearing capacity of 2,000 lbs/sf prior to installing foundation materials and/or AquaCell

Notes:

A. AASHTO ratings follow standard classification of truck axle loading weights for live loads.

B. Reference **AquaCell Cover Depths by Configuration (AQ-100-08.2)** and **AquaCell Installation Guide** for further information and instructions related to installation depths.

C. **AquaCell Extra Strong (EX)** consists of two base units inset within each other. Reference **TB101 & TB102 Load Capacity Testing Results** for additional information.

D. On-site soil compaction minimum percentage of **Standard Proctor Density (SP)** per ASTM D698 and ASTM D1557 as applicable.

E. Reference **Construction Loads over AquaCell System (AQ-100-20)** and **AquaCell Installation Guide** for guidelines on compaction and machinery used during installation.

F. The minimum width of sidewall backfill is **12 inches** or large enough to accommodate selected compaction equipment, whichever is greater.

G. Reference **Backfill Materials Table (AQ-100-01)** for recommended materials, AASHTO designations.

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Title: AquaCell Installation Min. Loading & Depth Requirements

Date: 10/28/2025 Scale: NTS Drawing: AQ-100-02 Rev: 4

Drawn: HJD Approved: JS Sheet: 1 of 1

Concrete collar for HS-20/25 (actual dimensions specified by E.O.R.)

⑩A

Geotextile (min. 6 oz./yd²)

Stone Backfill

AquaCell

Undisturbed Sub-base*

Notes:

A. For **Retention/ Infiltration systems**, provide a geotextile fabric with a minimum permittivity of 1.4 sec⁻¹. This will provide rapid water flow through the fabric which is essential to maintain infiltration performance and reduce the risk of clogging over time.

B. Wrapping the **AquaCell** system with geotextile also provides protection against abrasion, puncture, and UV degradation before and during installation. **Geosynthetics provided by others.**

C. Geotextile is layered around entirety of stone backfill/ & completely wrapped after initial backfill/ cover over top of tank is complete. Final backfill installed to surface grade as designed by E.O.R.

D. Reference **AquaCell Installation Guide** for further information and instruction on geosynthetics installation.

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Title: AquaCell Infiltration Detail

Date: 12/1/2025 Scale: NTS Drawing: AQ-100-03.1 Rev: 3

Drawn: HJD Approved: JS Sheet: 1 of 1

12" Top Adapter (3085857)

①

②

③

16" Top Adapter (2476842000)

③

Notes:

12" | 16" Top Adapter (3085857 | 2476842000) available for installation of a vertical pipe for venting and/or inspection into **AquaCell** tank. All castings provided by **OTHERS**.

1. Install vertical **12" PVC SDR-35/26 GASKETED bell end pipe (ASTM D3034/ F477)** over riser of 12" Top Adapter (Traffic | Non-traffic).

2. Install vertical **12" PVC SDR-35/26 PLAIN end pipe (ASTM D3034)** over riser of 12" Top Adapter (Traffic | Non-traffic). Use 12" flexible coupler & stainless steel straps to secure pipe & adapter.

3. Install vertical **12" HDPE plain end pipe (ASTM F2306 | F2648)** in riser of 16" top adapter (**Non-traffic applications only**). Wrap fabric around 12" Top Adapter and wrap with corrosion-resistant tape

4. **Not Shown** - 18" SCH.40 plain end pipe w/ ID 16.81" (ASTM D2665) over riser of 16" Top Adapter (Traffic | Non-traffic).

5. Reference **AquaCell Installation Guide & AquaCell Compatible Pipe Connections (AQ-200-02)** for further information and instruction on vertical & lateral pipe connections.

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Title: AquaCell Top Adapter Connection Detail

Date: 6/24/2025 Scale: NTS Drawing: AQ-100-04 Rev: 3

Drawn: HJD Approved: JS Sheet: 1 of 1

